

CYBERSECURITY HOMELAB CASE STUDY

SMB Brute-Force Detection and Alerting with Splunk

Controlled attack simulation, Windows event analysis,
and SIEM detection engineering

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*Authorized testing performed only against systems owned and configured inside an isolated
Proxmox laboratory network.*

Executive Summary

This project developed and validated a Splunk detection for repeated failed SMB authentication attempts against a Windows Server domain controller. A Kali Linux attacker VM was deployed on an isolated Proxmox network and used for service discovery, manual SMB validation, and a controlled Hydra password-guessing simulation. Windows Security events from DC01 were forwarded into Splunk Enterprise for analysis.

The final detection identified five Event ID 4625 network-logon failures from the same source address within five minutes. It was converted into a scheduled, high-severity alert. During validation, Hydra submitted five incorrect guesses before the known disposable test password on the sixth attempt. Splunk attributed the failures to the Kali address, 10.10.10.40, and the alert triggered successfully.

Outcome: The end-to-end workflow succeeded: attack activity was generated, logged by Windows, forwarded to Splunk, detected with SPL, and escalated as a high-severity alert.

Project Objectives

- Deploy a Kali attacker VM without exposing the household network.
- Identify the SMB service on DC01 and validate standard-user access controls.
- Confirm that successful and failed SMB authentication events reach Splunk.
- Write detection logic for repeated failed network logins from one source.
- Validate the scheduled alert using a controlled Hydra simulation.

Scope and Ethics

All activity was limited to mattlab.local, an authorized personal homelab. Kali had no default gateway and was attached to the isolated vmbr1 network. The target account was created specifically for testing and was not granted administrative privileges. A short custom password list was used instead of a large public wordlist to keep the test controlled.

1. Lab Architecture

The project extended an existing Active Directory and Splunk environment running on an HP EliteDesk Proxmox host. The attack, target, and monitoring components were separated by role while remaining observable through the SIEM pipeline.

System	Role	Address / Network
Kali VM 104	Attacker and testing system	10.10.10.40 / vmbr1
DC01	AD DS, DNS, SMB target, event source	10.10.10.10 / vmbr1
Splunk Enterprise	Search, detection, and alerting	10.10.10.30; 10.0.0.190
Universal Forwarder	Forwards DC01 Security events	TCP 9997 to Splunk
Proxmox vmbr1	Isolated cybersecurity lab segment	10.10.10.0/24

Detection Flow

1. Kali submits SMB authentication attempts to DC01 on TCP port 445.
2. Windows records successful and failed network logons in the Security log.
3. The Splunk Universal Forwarder sends those events to Splunk Enterprise.
4. SPL groups failed logons by source address and five-minute time bucket.
5. A matching result triggers the high-severity scheduled alert.

Key distinction: Hydra generated the test activity, but the main project deliverable was the Splunk detection and alert that identified the behavior.

2. Kali VM Deployment and Isolation

A prebuilt Kali QEMU image was selected because Proxmox uses QEMU/KVM virtualization. Windows extraction repeatedly failed, including attempts directed to the NAS, so 7-Zip was used to extract the archive successfully.

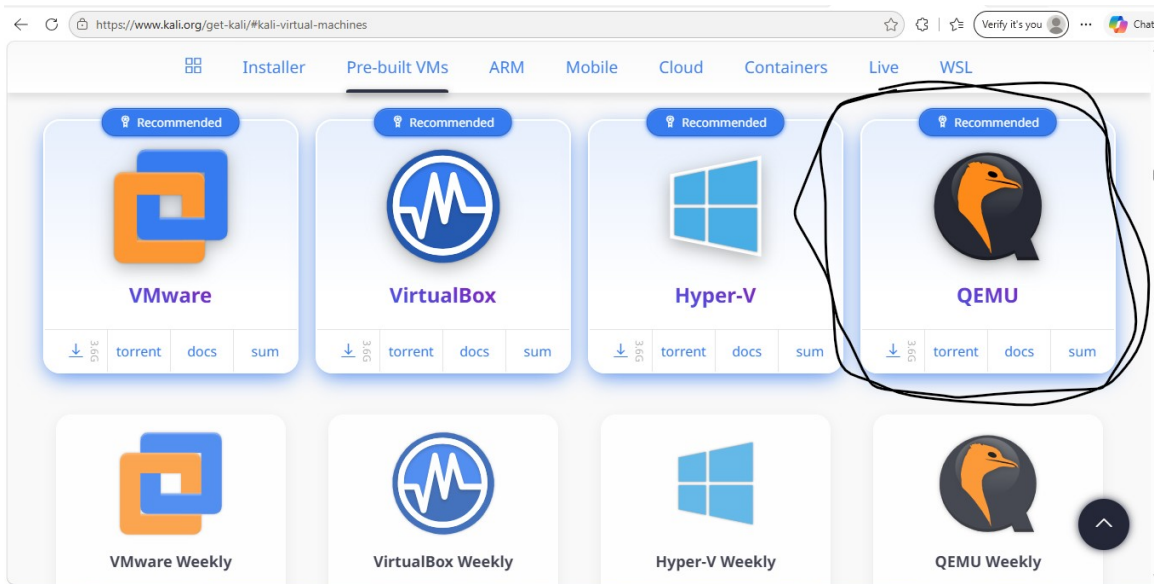


Figure 1. Kali QEMU virtual-machine image selected for the Proxmox deployment.

The extracted QCOW2 disk was transferred to the Proxmox host with SCP after the browser upload method failed. The image was verified, imported into local-lvm, attached to VM 104 as SCSI0, and booted.

```

matthew@evil-lebaron:~$ client_loop: send disconnect: Connection reset
PS C:\Users\STUDENT LOANER> ssh matthew@10.0.0.216
matthew@10.0.0.216's password:
Linux evil-lebaron 6.12.75+rpt-rpi-v8 #1 SMP PREEMPT Debian 1:6.12.75-1+rpt1 (2026-03-11) aarch64

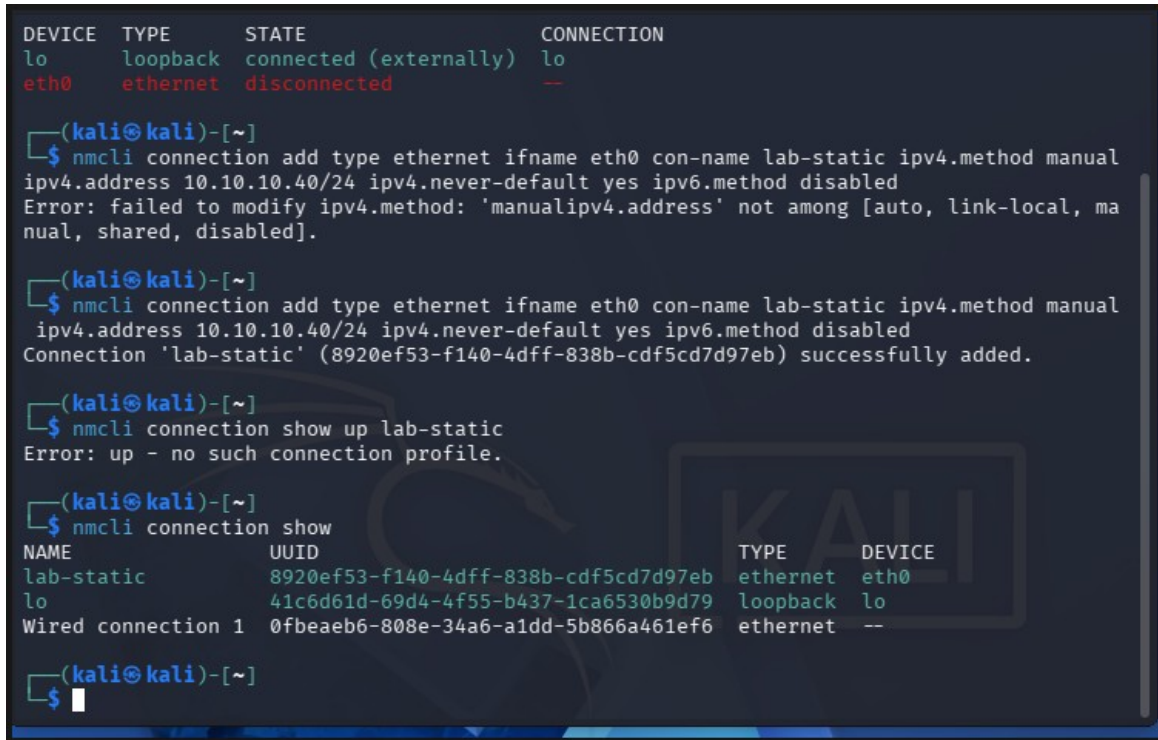
The programs included with the Debian GNU/Linux system are free software;
the exact distribution terms for each program are described in the
individual files in /usr/share/doc/*/copyright.

Debian GNU/Linux comes with ABSOLUTELY NO WARRANTY, to the extent
permitted by applicable law.
Last login: Wed Sep  2 20:24:44 2026 from 10.0.0.32
matthew@evil-lebaron:~$ sudo find "/srv/dev-disk-by-uuid-87534542-ff06-43cf-9b22-c40503850684/matts_nas/VMs/Kali" -maxdepth 3 -type f \( -name "*.q
cow2" -o -name "*.iso" \) -printf '%p | %s bytes\n'
[sudo] password for matthew:
/srv/dev-disk-by-uuid-87534542-ff06-43cf-9b22-c40503850684/matts_nas/VMs/Kali/kali-linux-2026.2-qemu-amd64/kali-linux-2026.2-qemu-amd64.iso | 159996
31360 bytes
matthew@evil-lebaron:~$ scp "/srv/dev-disk-by-uuid-87534542-ff06-43cf-9b22-c40503850684/matts_nas/VMs/Kali/kali-linux-2026.2-qemu-amd64/kali-linux-
2026.2-qemu-amd64.iso" root@10.0.0.249:/var/lib/vz/template/iso/kali-linux-2026.2-qemu-amd64.qcow2
The authenticity of host '10.0.0.249 (10.0.0.249)' can't be established.
ED25519 key fingerprint is SHA256:by8uteTPLWspNMM8plgHcorJges8xb8x8q0MhokPcE.
This key is not known by any other names.
Are you sure you want to continue connecting (yes/no/[fingerprint])? yes
Warning: Permanently added '10.0.0.249' (ED25519) to the list of known hosts.
root@10.0.0.249's password:
kali-linux-2026.2-qemu-amd64.iso
matthew@evil-lebaron:~$
  
```

Figure 2. Kali QCOW2 disk transferred to the Proxmox host through SCP.

Network Configuration

Kali received the static address 10.10.10.40/24 on vmbr1. No default gateway was configured. This allowed communication with lab systems while preventing a route to the household router or the Internet.



```

DEVICE  TYPE      STATE          CONNECTION
lo      loopback  connected (externally) lo
eth0    ethernet  disconnected    --

(kali㉿kali)-[~]
$ nmcli connection add type ethernet ifname eth0 con-name lab-static ipv4.method manual
ipv4.address 10.10.10.40/24 ipv4.never-default yes ipv6.method disabled
Error: failed to modify ipv4.method: 'manual' not among [auto, link-local, manual, shared, disabled].

(kali㉿kali)-[~]
$ nmcli connection add type ethernet ifname eth0 con-name lab-static ipv4.method manual
ipv4.address 10.10.10.40/24 ipv4.never-default yes ipv6.method disabled
Connection 'lab-static' (8920ef53-f140-4dff-838b-cdf5cd7d97eb) successfully added.

(kali㉿kali)-[~]
$ nmcli connection show up lab-static
Error: up - no such connection profile.

(kali㉿kali)-[~]
$ nmcli connection show
NAME                UUID                                  TYPE      DEVICE
lab-static          8920ef53-f140-4dff-838b-cdf5cd7d97eb ethernet  eth0
lo                  41c6d61d-69d4-4f55-b437-1ca6530b9d79 loopback  lo
Wired connection 1  0fbaeb6-808e-34a6-a1dd-5b866a461ef6 ethernet  --
  
```

Figure 3. Kali static connection profile configured for 10.10.10.40/24 without a default gateway.

Isolation result: Kali could communicate with DC01 at 10.10.10.10 but could not reach the household router at 10.0.0.1. This confirmed that the attack VM remained inside the intended lab boundary.

Deployment Challenges

- Insufficient Windows storage prevented local extraction of the original archive.
- NAS-based extraction failed after long transfer attempts.
- The Proxmox browser upload did not accept the QCOW2 workflow reliably.
- 7-Zip and SCP provided successful alternatives.

3. Service Discovery and Test Account

Nmap Reconnaissance

A targeted Nmap service scan checked common administrative and web ports on DC01. TCP port 445 was the only tested port reported open and was identified as Microsoft SMB. Ports 21, 22, 80, 443, and 3389 were filtered.

```
nmap -sV -p 21,22,80,443,445,3389 10.10.10.10
```

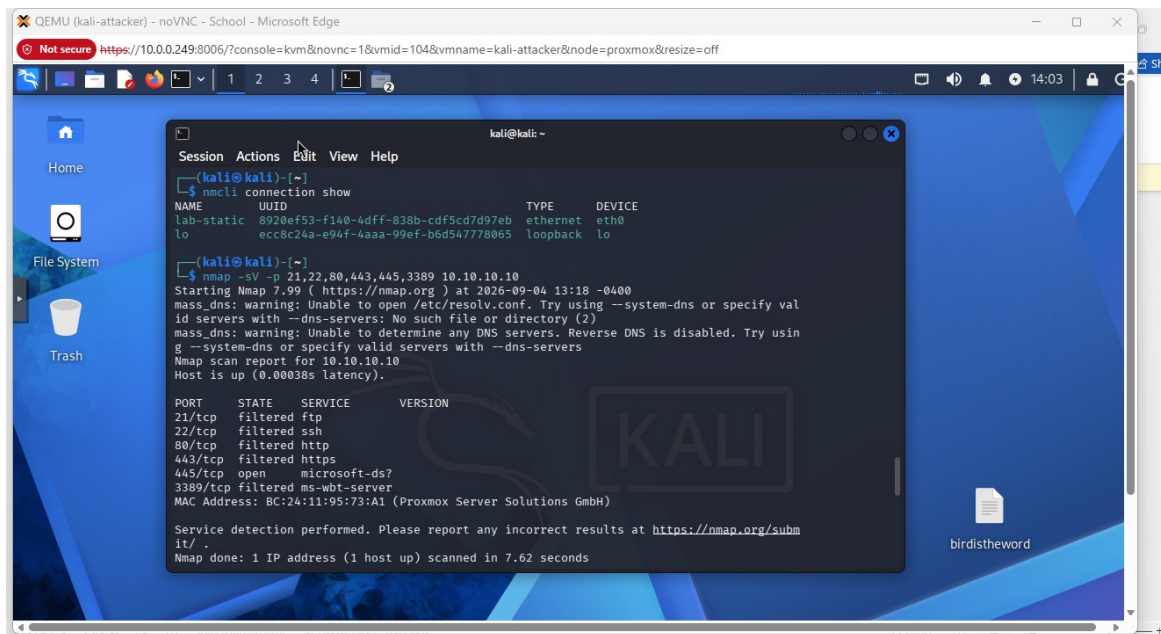


Figure 4. Nmap identifies TCP port 445 as the available SMB attack surface.

DNS warnings in the scan were expected because the isolated Kali VM had no DNS server or default route. They did not prevent the TCP scan from completing.

Disposable Active Directory Account

A standard domain account named `smb_test` was created specifically for authentication testing. The account was intentionally excluded from `IT_Admins` and `Domain Admins` so that authentication and authorization could be evaluated separately.

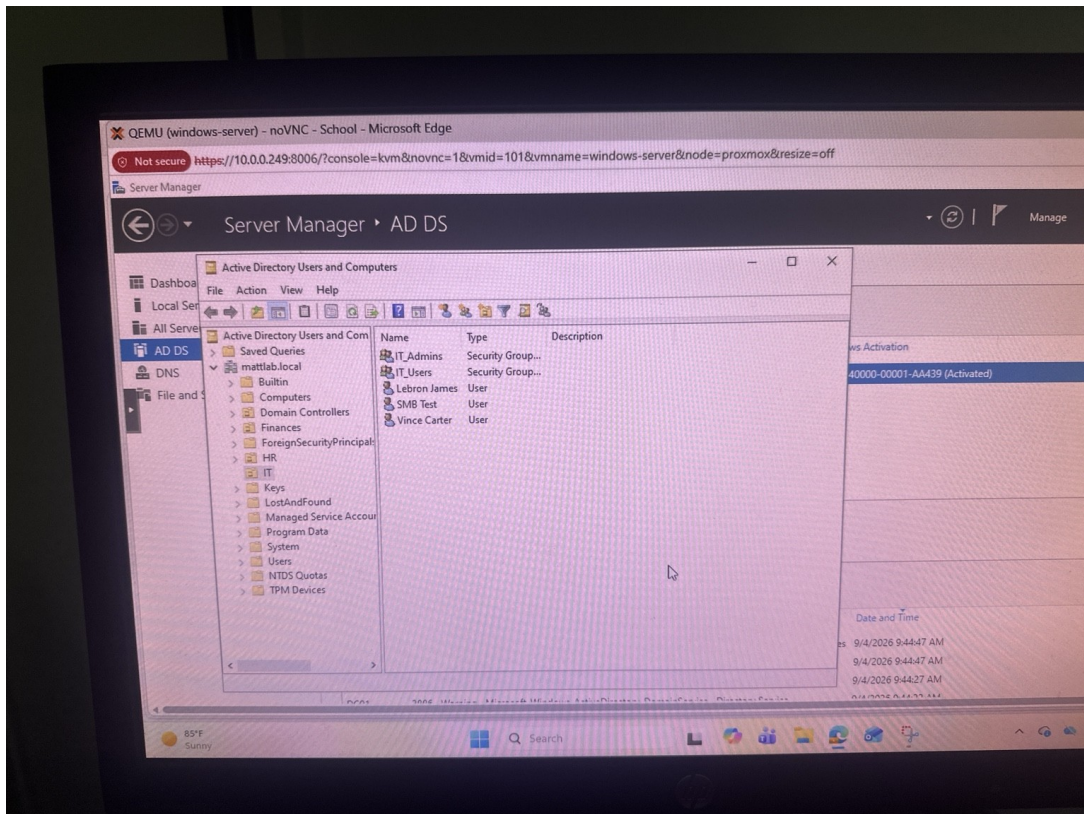


Figure 5. Disposable SMB Test account created in Active Directory Users and Computers.

4. SMB Authentication and Access-Control Validation

Successful Authentication

The smbclient utility authenticated to DC01 with the disposable domain credentials and requested the available shares. The response listed ADMIN\$, C\$, IPC\$, NETLOGON, and SYSVOL, confirming that SMB authentication and enumeration were working.

```
smbclient -L //10.10.10.10 -U 'MATTLAB\smb_test'
```

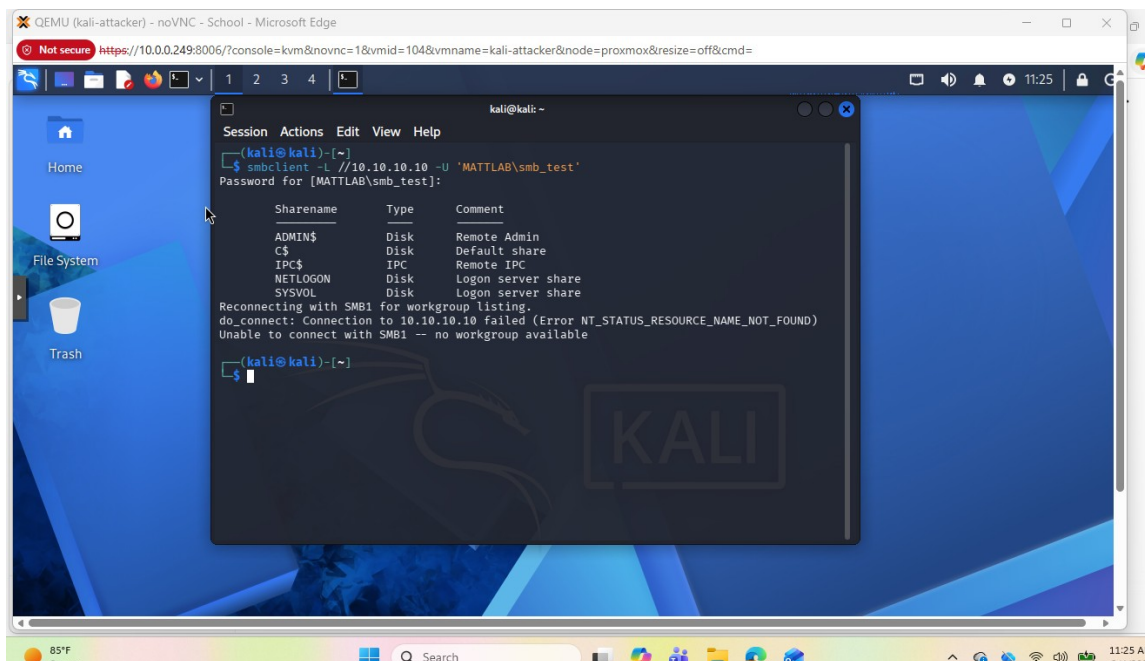


Figure 6. Successful SMB authentication and share enumeration from Kali.

Authorization Denial

The same valid credentials were then used to request the protected ADMIN\$ share. DC01 returned NT_STATUS_ACCESS_DENIED. This proved that the account could authenticate while remaining blocked from administrative resources.

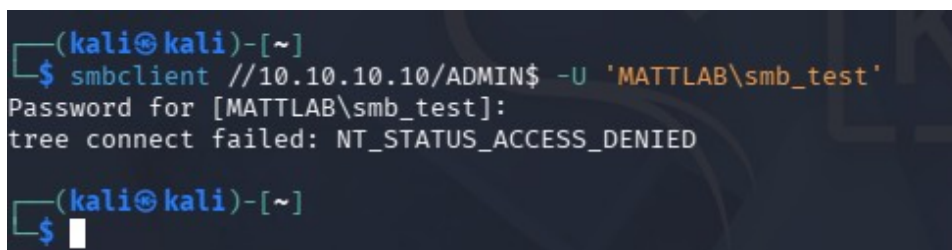


Figure 7. The standard account is denied access to the ADMIN\$ administrative share.

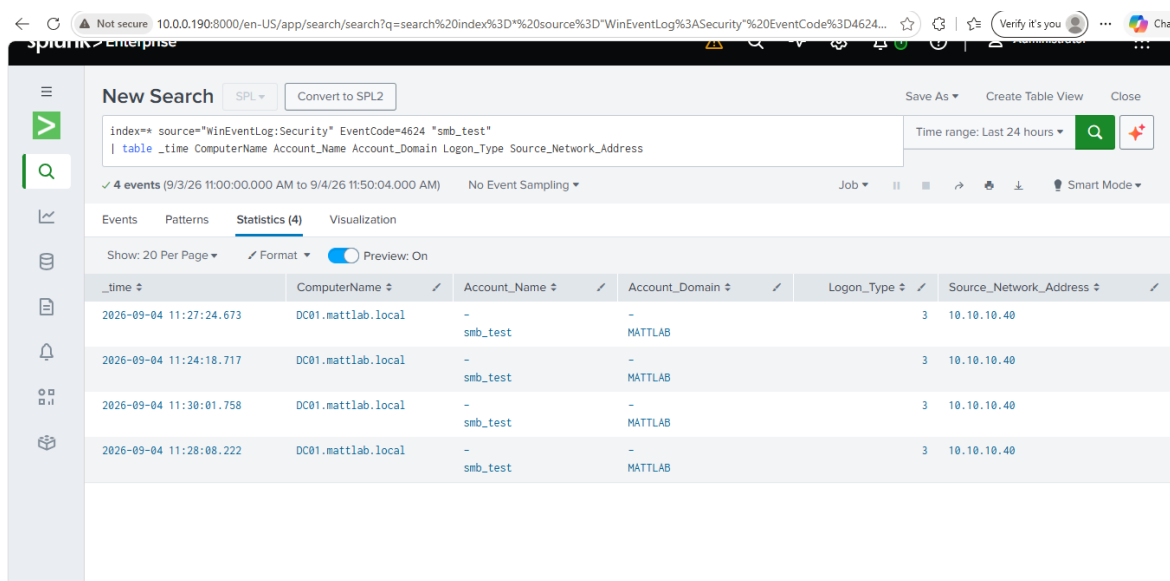
Security principle: Authentication answered who the user was; authorization determined that the user was not permitted to enter ADMIN\$. The result demonstrated least-privilege enforcement.

5. Windows Event and Splunk Validation

Manual correct and incorrect SMB logins were generated before the alert was built. This established that the complete logging pipeline was functional and that the relevant event fields were available for detection.

Successful Network Logons - Event ID 4624

Successful SMB authentication produced Event ID 4624 with Logon Type 3. The event identified DC01 as the destination, MATTLAB\smb_test as the account, and 10.10.10.40 as the source address.



_time	ComputerName	Account_Name	Account_Domain	Logon_Type	Source_Network_Address
2026-09-04 11:27:24.673	DC01.mattlab.local	smb_test	MATTLAB	3	10.10.10.40
2026-09-04 11:24:18.717	DC01.mattlab.local	smb_test	MATTLAB	3	10.10.10.40
2026-09-04 11:30:01.758	DC01.mattlab.local	smb_test	MATTLAB	3	10.10.10.40
2026-09-04 11:28:08.222	DC01.mattlab.local	smb_test	MATTLAB	3	10.10.10.40

Figure 8. Splunk displays successful SMB network logons from the Kali source address.

Failed Network Logons - Event ID 4625

Incorrect credentials produced Event ID 4625 with Logon Type 3 and the failure reason “Unknown user name or bad password.” Splunk again attributed the activity to 10.10.10.40.

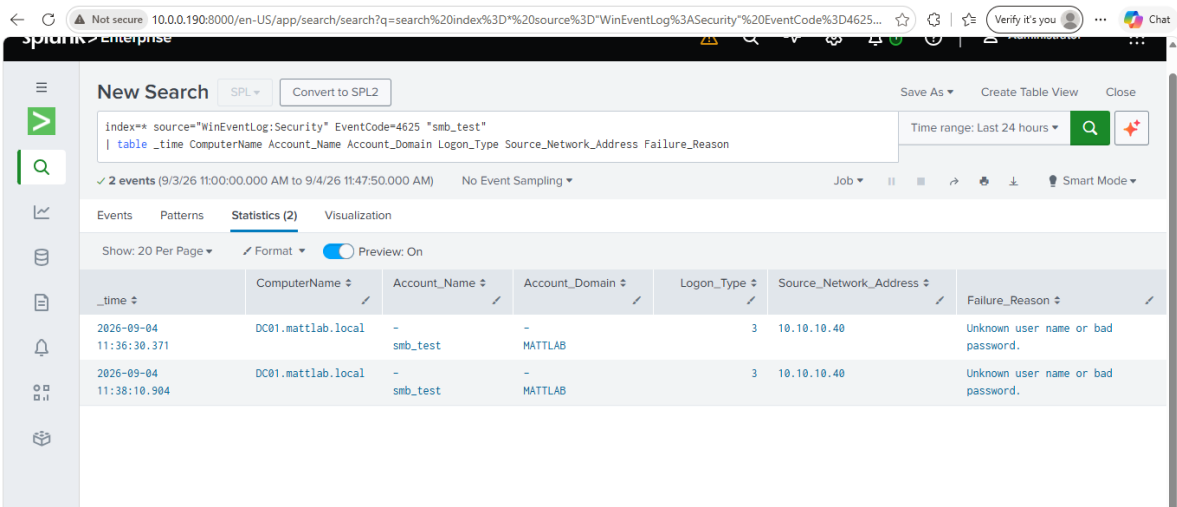


Figure 9. Splunk displays failed SMB network logons from Kali with the account and failure reason.

Observed Detection Fields

Field	Observed value	Analytical value
EventCode	4625	Identifies a failed Windows logon.
Logon_Type	3	Ties the failure to network authentication such as SMB.
Account_Name	smb_test	Identifies the targeted disposable account.
Source_Network_A ddress	10.10.10.40	Attributes the activity to the Kali VM.
Failure_Reason	Unknown user name or bad password	Provides context for triage and threshold tuning.

Pipeline checkpoint: At this point the source, destination, account, outcome, logon type, and failure reason were all searchable in Splunk. The telemetry was sufficient to build a source-based threshold alert.

6. Detection Engineering

The detection was designed to avoid alerting on every isolated password mistake. Instead, it identifies five or more Event ID 4625 network logons from the same source address within a five-minute bucket.

SPL Detection

```
index=* source="WinEventLog:Security" EventCode=4625 Logon_Type=3
| bin _time span=5m
| stats count AS failed_attempts values(Account_Name) AS targeted_accounts values(Failure_Reason) AS
failure_reasons BY _time Source_Network_Address
| where failed_attempts >= 5
```

Detection element	Purpose
EventCode=4625	Restricts results to failed Windows logons.
Logon_Type=3	Focuses on network authentication such as SMB.
bin _time span=5m	Places related activity into five-minute windows.
stats ... BY source	Counts failures and preserves account and reason context.
failed_attempts >= 5	Returns only activity that crosses the alert threshold.

Alert Configuration

The search was saved as Repeated SMB Failed Logins and configured as a scheduled search using the cron expression `*/5 * * * *`. It ran every five minutes over the most recent five-minute search window. A result count greater than zero triggered the alert once per scheduled run. The alert action added the event to Triggered Alerts with High severity.

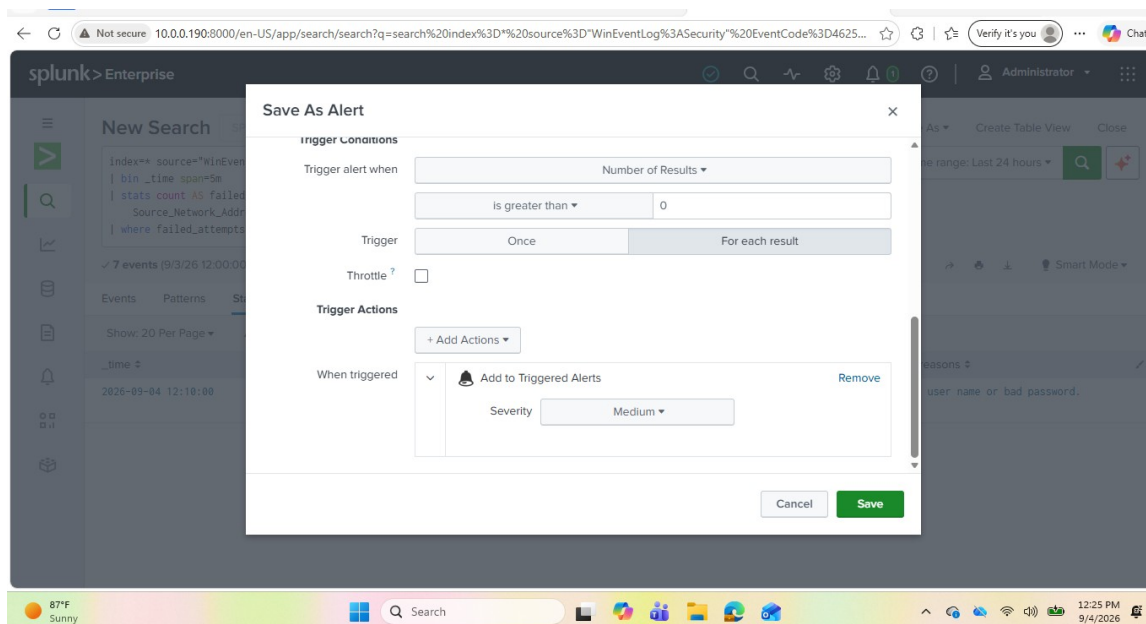


Figure 10. Alert condition and action during configuration; the final alert was saved with Once and High severity.

7. Controlled Hydra Attack Simulation

After validating the detection manually, Hydra was used to automate SMB password guesses against the disposable account. A custom six-entry password list kept the test controlled: five incorrect guesses were placed before the known temporary password on the sixth line.

This was a simulation rather than an attempt to discover an unknown credential. Its purpose was to reproduce the event pattern that the Splunk alert was designed to detect.

```
hydra -l smb_test -P ~/Desktop/birdistheword -m 'workgroup:{MATTLAB}' -V -f smb2://10.10.10.10
```

Hydra Result

Hydra submitted all six credentials and reported the valid pair on the sixth attempt. The temporary password has been redacted from the evidence image. The SMB2 module also produced a warning about possible credential acceptance ambiguity, so the Hydra success line was not treated as proof by itself. Earlier smbclient testing had already validated the known credentials, while Splunk independently verified the five failed attempts.

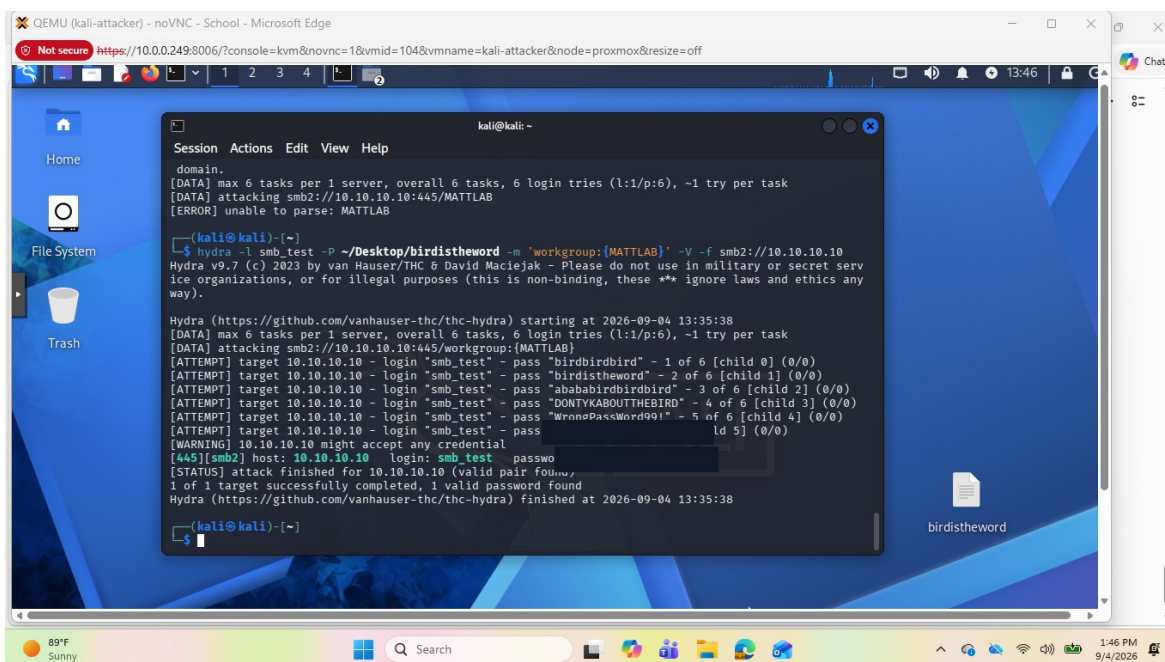


Figure 11. Hydra completes the controlled six-entry SMB2 password-guessing simulation. Temporary credential redacted.

Authenticity: The short wordlist made the lab repeatable. The authentic project outcome was the observable attack pattern and the successful detection—not the difficulty of guessing a password that was intentionally included.

8. Alert Validation and Results

The scheduled detection grouped the Hydra failures into the 1:35 PM five-minute window. It returned five failed attempts from 10.10.10.40 against smb_test with the expected bad-password failure reason.

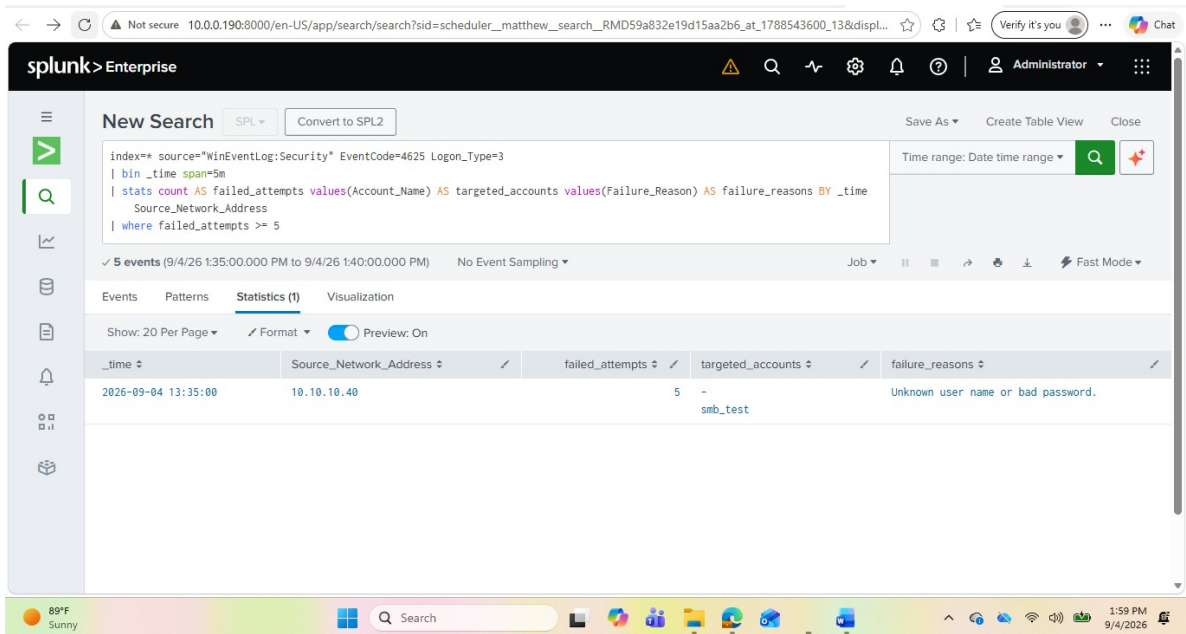


Figure 12. Detection result showing five failed attempts from Kali during one five-minute bucket.

At the next scheduled execution, Splunk created a second Repeated SMB Failed Logins alert at 1:40 PM. The earlier 12:55 PM alert came from the manual validation traffic; the 1:40 PM alert came from the Hydra simulation.

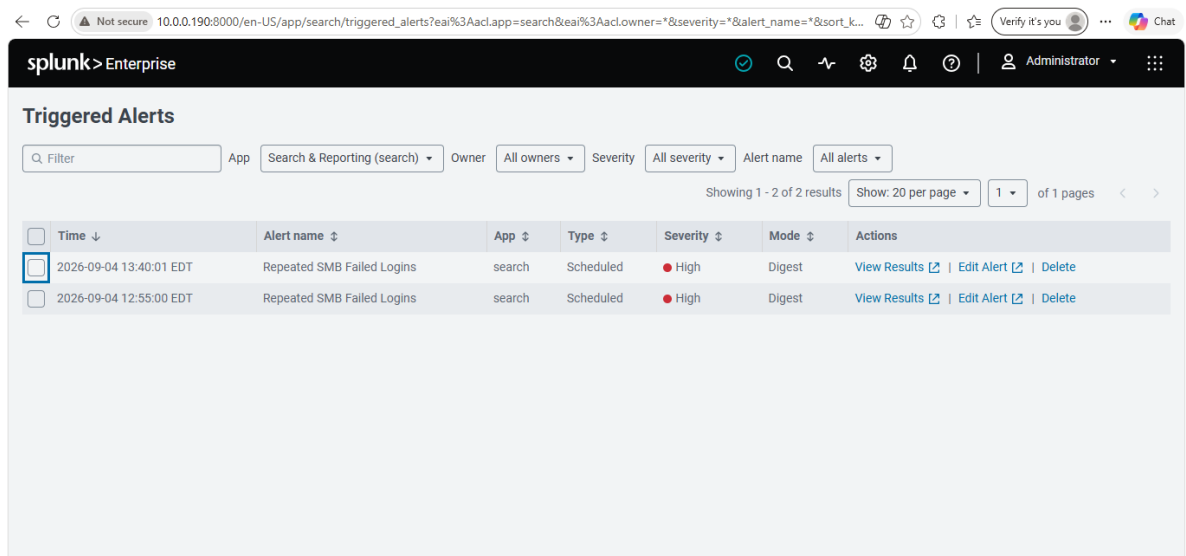


Figure 13. High-severity scheduled alert triggered after the Hydra simulation.

Validated Outcome

Control	Evidence	Result
Network isolation	No default gateway on Kali	Passed
Service discovery	TCP 445 open on DC01	Passed
Least privilege	ADMIN\$ returned access denied	Passed
Event collection	4624 and 4625 searchable	Passed
Detection logic	Five failures grouped by source	Passed
Alerting	High-severity alert at 1:40 PM	Passed

9. Security Analysis and Recommendations

Findings

- SMB was reachable from the isolated attacker segment and accepted domain authentication.
- The standard account could enumerate common shares but could not access ADMIN\$.
- Windows audit events contained the source IP, account, logon type, and failure reason needed for investigation.
- Five rapid failures were distinguishable from a single ordinary password mistake.
- The alert detected the automated pattern before the final successful credential in the controlled list.

Recommended Defensive Actions

- Restrict TCP 445 to systems and network segments that require SMB access.
- Use an account-lockout policy appropriate to the organization's risk and support requirements.
- Investigate repeated 4625 events by source address, account, and subsequent successful logons.
- Correlate repeated failures with Event ID 4624 to identify possible successful compromise after guessing.
- Use strong, unique passwords and multifactor authentication where the service supports it.
- Tune the five-attempt threshold against normal organizational behavior to reduce false positives.

Response Workflow

- Validate the source IP and determine whether it belongs to an approved system.
- Identify targeted accounts and confirm whether any later authentication succeeded.
- Contain suspicious sources or accounts when the activity is unauthorized.
- Reset exposed credentials, review related activity, and preserve relevant logs.
- Tune the alert after reviewing confirmed incidents and benign patterns.

10. Challenges and Lessons Learned

Challenge	Resolution / lesson
Archive extraction failed locally and over the NAS	Used 7-Zip and changed the extraction workflow.
Proxmox browser upload failed for QCOW2	Transferred the disk through SCP and imported it into local-lvm.
Initial command was entered on the wrong VM	Verified the active system and reran Nmap from Kali.
Hydra could not find the wordlist	Used the correct absolute Desktop path.
Hydra rejected the MATTLAB option	Used the SMB2 workgroup:{MATTLAB} module syntax.
The alert did not appear immediately	Accounted for forwarding, indexing, and the five-minute schedule.

Technical Lessons

- A 4624 event proves successful authentication; it does not automatically prove authorization to a protected resource.
- Logon Type 3 is an important field for distinguishing network authentication from interactive logons.
- A detection threshold is a tradeoff: a lower value detects earlier but may create more noise.
- Scheduled alerts introduce a short delay because data must be collected, indexed, and evaluated.
- Tool output should be validated against independent telemetry, especially when a module reports possible false positives.

Cleanup

The smb_test account and local password-list file were created only for this exercise and will be deleted afterward. The Kali VM will also be restored to its pre-test snapshot for future projects.

Conclusion

This project successfully demonstrated an end-to-end attack-simulation and detection-engineering workflow. Kali generated controlled SMB authentication activity against DC01, Windows recorded the activity, the Universal Forwarder delivered the events to Splunk, and custom SPL identified the repeated failures. The resulting scheduled alert triggered at High severity after the Hydra simulation.

The project combined virtualization, network isolation, Active Directory, Linux tooling, Windows auditing, SIEM administration, and detection validation. Most importantly, it showed that defensive logic can be tested with observable evidence instead of being assumed to work.

Skills Demonstrated

- Proxmox VM deployment and isolated network design
- Linux network configuration and troubleshooting
- Nmap service discovery and SMB validation
- Active Directory test-account administration
- Windows Security Event ID 4624/4625 analysis
- Splunk SPL development and scheduled alert configuration
- Controlled Hydra password-guessing simulation
- Evidence-based troubleshooting and security documentation

Final result: Hydra simulation -> five Windows logon failures -> Splunk correlation -> high-severity alert.